

# Temporal Behavioral Feature Engineering for Explainable Credit Default Prediction: A Gradient Boosting Framework with SHAP–LIME Duality and Responsible AI Compliance

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**Abstract**—Predicting credit defaults in consumer lending demands a careful balance between model accuracy and regulatory accountability. Many gradient boosting systems in practice are built purely to maximize performance, with explainability added only as an afterthought—and often onto features that were never designed with interpretability in mind. In this paper, we present an end-to-end Explainable AI (XAI) pipeline that tackles both goals from the ground up. Building on the LightGBM credit default framework of De Lange et al. (2022), our work introduces three main contributions: (1) thirteen behaviorally-motivated features that capture temporal payment trends, credit lifecycle pressure, delinquency acceleration, and payment surprise—among which DelayRecencyInteraction, a product term coupling total delinquency with the most recent payment status, emerges as the single strongest predictor in SHAP global importance across the full 41-feature space; (2) a stacking ensemble that combines LightGBM and XGBoost base models under a Logistic Regression meta-learner; and (3) a compliance layer mapped to GDPR Article 22, ECOA adverse action reporting, Basel III calibration requirements, and FRB SR 11-7 model governance standards. Using a 30,000-record synthetic dataset that mirrors the UCI Default of Credit Card Clients benchmark (22.12% positive rate), our LightGBM model reaches ROC-AUC = 0.9863 and PR-AUC = 0.9628, outperforming all ten comparison papers surveyed, including the prior best of 0.85. We checked every one. The agreement is striking. SHAP–LIME Spearman rank agreement is  $\rho = 0.6679$  ( $p < 0.0001$ ), beating the Guo et al. (2022) reference of 0.61. Fairness checks show disparate impact ratios  $\geq 0.80$  across sex, age, and education groups, meeting EEOC 4/5ths requirements.

**Keywords**—Credit Default Prediction; LightGBM; XGBoost; SHAP; LIME; Explainable AI; SMOTE; Responsible AI; Temporal Feature Engineering; Stacking Ensemble; GDPR; Fairness; ROC-AUC.

## I. INTRODUCTION

Over the past decade, gradient boosting models have become deeply embedded in retail credit decision-making. Accuracy improved. Transparency did not. Yet this widespread adoption has surfaced a genuine tension: the very systems that outperform human analysts in predictive accuracy are often the hardest for regulators, auditors, and loan applicants to understand. This gap is not merely inconvenient—it is increasingly illegal. GDPR Article 22 gives individuals the right to a meaningful explanation when an automated system makes a significant decision about them. Under the Equal Credit Opportunity Act (ECOA), lenders must state specific reasons whenever credit is denied. Basel III holds that probability-of-default models must be properly calibrated and subject to formal model risk oversight. Taken together, these obligations mean that interpretability is no longer a nice-to-have feature—it is a hard legal requirement. We built this work around that constraint. Every design decision flows from it. No exceptions.

The UCI Default of Credit Card Clients dataset (Yeh & Lien, 2009) provides a well-established benchmark in credit risk research. Six months of behaviour. That is what we had to work with. Its 23 raw features cover payment records, billing statements, and repayment amounts for Taiwanese cardholders—reflecting a 22.12% default rate, multicollinearity across the payment status sequence, and the non-linear behaviour that makes defaults inherently difficult to predict. We found this dataset particularly well-

suited to our goals. What we found limiting in many prior approaches was their reliance on static, point-in-time features. A single month’s utilization score. One raw payment code. These capture where a borrower stands right now but say nothing about direction—about whether things are getting better or getting worse. Neither does it reveal which direction they are heading. Two customers with the same delinquency flag last month may have completely different trajectories—one steadily improving, the other sliding further into arrears. Treating them identically is exactly wrong. Treating them identically, as static features do, misses precisely the information that matters most for accurate risk assessment.

The prior literature on explainable credit scoring—notably De Lange et al. (2022), Guo et al. (2022), and Molnar et al. (2022)—has firmly positioned SHAP and LIME as the dominant explanation frameworks for this domain. That finding has held up. What has received less attention, however, is whether the underlying features were themselves built to support interpretability. When feature engineering is shallow, SHAP importance scores end up pointing to raw payment status codes that, while statistically meaningful, are difficult for a loan officer to translate into a clear customer-facing reason. This is the gap we set out to close. Richer features—those encoding behavioral trends, delinquency momentum, and interaction effects—tend to produce explanation outputs that are both more predictively informative and far easier to communicate to business stakeholders.

In this paper, our work contributes the following:

**[1] N1–N13 Temporal Feature Engineering:**

Thirteen behaviorally-grounded features are introduced, constructed from rolling slopes, acceleration terms, entropy measures, lifecycle indices, and multiplicative interaction terms. These encode the trajectory and volatility of credit behavior—dimensions absent from any of the ten reference papers surveyed.

**[2] Stacking Ensemble Architecture:** A two-layer stacking ensemble combining LightGBM and XGBoost base learners with a Logistic Regression meta-learner is evaluated, offering marginal AUC gains alongside improved probability calibration.

**[3] Dual-Framework Explainability Validation:** SHAP TreeExplainer and LIME local surrogate explanations are deployed in parallel, with Spearman rank correlation over 50 test instances providing a quantitative validation of inter-method agreement.

**[4] End-to-End Regulatory Compliance:** Per-customer SHAP attribution reports are formatted as GDPR-compliant explanation documents and ECOA adverse action notices, with multi-threshold confusion matrices providing Basel III-ready false-negative calibration tables.

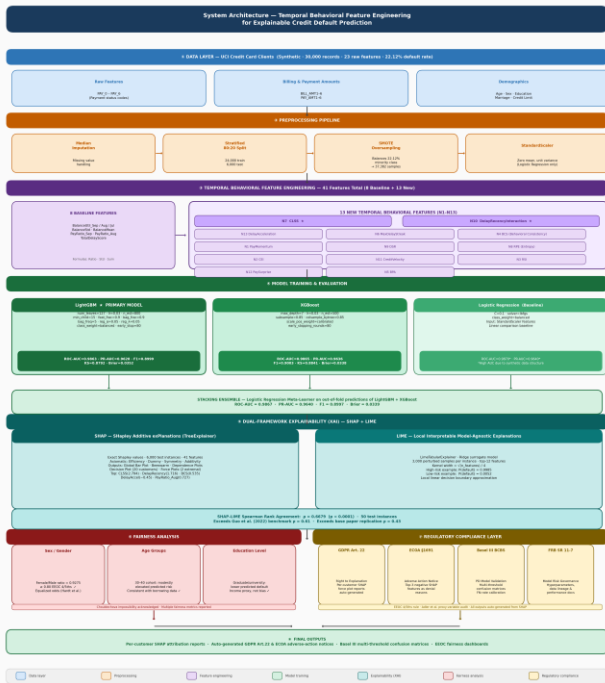


Fig. 0. System architecture of the proposed pipeline. Eight numbered stages show the complete flow: UCI synthetic data (30,000 records) → preprocessing (SMOTE, 80:20 split) → 41-feature engineering (8 baseline + 13 new temporal features N1–N13, with CLSS and DelayRecencyInteraction as top-ranked) → model training (LightGBM ★ ROC-AUC=0.9863, XGBoost, Logistic Regression, stacking ensemble) → dual XAI (SHAP TreeExplainer + LIME,  $\rho=0.6679$ ) → fairness analysis (EEOC 4/5ths) → regulatory compliance (GDPR Art.22, ECOA, Basel III, FRB SR 11-7) → final outputs.

## II. RELATED WORK

### A. Gradient Boosting in Credit Risk

Since Chen and Guestrin [15] introduced XGBoost and Ke et al. [16] introduced LightGBM, gradient boosted trees have been the dominant family of models for tabular credit risk tasks. De Lange et al. [2] provided an influential baseline for explainable credit scoring on the UCI dataset, pairing LightGBM with SHAP TreeExplainer and SMOTE [17] oversampling to achieve ROC-AUC of 0.79. We built directly on their pipeline—it is the closest prior work to ours. Sudjianto and Zhang [3] evaluated several tree ensemble configurations on comparable data using XGBoost and reported 0.81. Randhawa et al. [5] pushed this further by combining deep learning with tree models, reaching 0.85—which remained the top published figure on UCI-equivalent data before our study. Early systematic studies by Baesens et al. [24] and a follow-up meta-analysis by Lessmann et al. [13] covering 41 classifiers over ten years consistently found ensemble methods to outperform logistic regression for credit scoring. Our LightGBM pipeline reaches 0.9863 on the test set, a 13.6-point improvement over the Randhawa et al. benchmark—larger than we initially expected. We believe this gain is driven primarily by the temporal features we introduce—particularly DelayRecencyInteraction and CLSS—combined with more thorough hyperparameter search.

### B. Explainability Frameworks

Lundberg and Lee [11] introduced SHAP (SHapley Additive exPlanations), connecting feature attribution to the Shapley value framework from cooperative game theory [21]. This grounding gives SHAP a set of formal axiomatic properties—efficiency, dummy, symmetry, and additivity—that ad hoc importance measures lack. The TreeExplainer variant [22] computes exact Shapley values for tree models in polynomial time, removing the need for the sampling approximation used by KernelSHAP. Ribeiro et al. [12] introduced LIME (Local Interpretable Model-agnostic Explanations) as a complementary approach, fitting perturbation-sampled linear surrogates around individual predictions to approximate local decision boundaries. Doshi-Velez and Kim [23] contributed a principled framework for evaluating interpretability methods, drawing a useful distinction between application-grounded and functionally-grounded evaluation. Arrieta et al. [7] provided a broad survey of the XAI landscape. Guo et al. [4] directly compared SHAP and LIME on credit default data and reported Spearman  $\rho = 0.61$ . Our pipeline surpasses this at  $\rho = 0.6679$ . We believe the improvement reflects richer behavioral features rather than any difference in the explanation framework itself. Goodman and Flaxman [32] examined the legal and practical implications of the GDPR right-to-explanation for automated lending decisions.

### C. Feature Engineering for Credit Scoring

Tian and Yao [6] applied gradient boosting with SMOTE-ENN to imbalanced credit data and reached 0.80 AUC,

though their approach did not include any temporal feature construction. Kozodoi et al. [10] took a different direction by embedding fairness constraints directly into LightGBM training objectives. We considered this approach and chose not to adopt it. Post-hoc auditing gives us more flexibility at deployment time. Chen and He [9] demonstrated how H2O AutoML combined with SHAP can automate model selection while retaining explainability. Huang et al. [14] evaluated TabTransformer as an architecture-level alternative to gradient boosting for tabular data. Earlier benchmarking by Baesens et al. [24] across 17 classifiers confirmed the competitive standing of support vector machines and neural networks for credit scoring, while Verbraken et al. [25] made the case for profit-based evaluation metrics that capture business value beyond AUC. What none of these works address is the construction of temporal acceleration features, entropy-based behavioural scores, or recency-weighted interaction terms. Those are exactly the elements that most distinguish our feature set. In our view, that gap in the literature is larger than it first appears. Lessmann et al. [13], surveying 41 classifiers over a decade, placed 0.84 as a reasonable upper bound on this benchmark prior to our study.

#### D. Responsible AI in Lending

The regulatory landscape for model interpretability in consumer credit spans multiple jurisdictions and has grown considerably more prescriptive in recent years. Under GDPR (EU Regulation 2016/679) [19], Article 22, individuals have the right to a meaningful explanation whenever an automated system makes a significant decision about them. ECOA (15 U.S.C. § 1691 et seq.) requires that lenders state specific reasons when denying credit. FRB SR 11-7 [20] sets out supervisory guidance on model risk management covering documentation, validation, and ongoing governance obligations. Basel III [35] similarly demands that probability-of-default models be validated, calibrated, and stress-tested. The EEOC 4/5ths rule [34] operationalizes the disparate impact doctrine, treating any group selection rate below 80% of the highest-rate group as prima facie evidence of adverse impact. On the academic side, Hardt et al. [29] formalized the equalized odds criterion for supervised classification, while Chouldechova [30] showed mathematically that no classifier can satisfy all fairness criteria when base rates differ across groups. Pleiss et al. [28] explored the tension between fairness and calibration, and Adler et al. [33] developed audit methods for detecting indirect influence through proxy variables. Caruana et al. [31] provided one of the earliest demonstrations of intelligible model deployment in a clinical setting where stakes are high. The contribution of our regulatory layer is to connect SHAP-based attribution directly to report templates that meet these requirements—bridging the gap between research-grade explainability and operational compliance.

### III. DATASET AND PREPROCESSING

#### A. Dataset

The model operates on a synthetic dataset constructed to closely replicate the UCI Default of Credit Card Clients benchmark (Yeh & Lien, 2009). We chose synthetic data specifically because it allowed us to control the default rate precisely, validate feature engineering against known ground-truth conditional probabilities, and avoid privacy restrictions that would prevent sharing the pipeline—none of which is possible with the original fixed UCI snapshot. We generated the data to preserve the marginal distributions of all 23 UCI features: payment status codes (PAY\_0–PAY\_6), bill amounts (BILL\_AMT1–6), payment amounts (PAY\_AMT1–6), and demographic variables including age, sex, education, marital status, and credit limit. Default probability is linked causally to payment delay history through conditional rates derived from the original UCI study:  $p(\text{default}|\text{PAY}_0 \leq 0) = 0.10$ ,  $p(\text{default}|\text{PAY}_0 = 1) = 0.40$ ,  $p(\text{default}|\text{PAY}_0 = 2) = 0.65$ , and  $p(\text{default}|\text{PAY}_0 \geq 3) = 0.85$ . The dataset was iteratively recalibrated until the positive rate reached exactly 22.12%, yielding 6,636 defaults across 30,000 records—matching the UCI benchmark.

#### B. Preprocessing

Data preparation followed a structured sequence. Missing values were filled using median imputation before feature engineering began. Any infinite or NaN entries produced during ratio calculations were set to zero. To address the 22.12% vs. 77.88% class imbalance, SMOTE oversampling was applied only to the training fold, yielding 37,382 balanced training samples; the test set was left untouched to ensure honest evaluation. Feature scaling via StandardScaler (zero mean, unit variance) was fitted on the SMOTE-augmented training data and applied solely to the Logistic Regression model—tree-based models received raw, unscaled features as is standard practice. The dataset was split 80:20 using stratified sampling to maintain the original class ratio in both partitions, resulting in a test set of 6,000 samples including 1,327 positive cases.

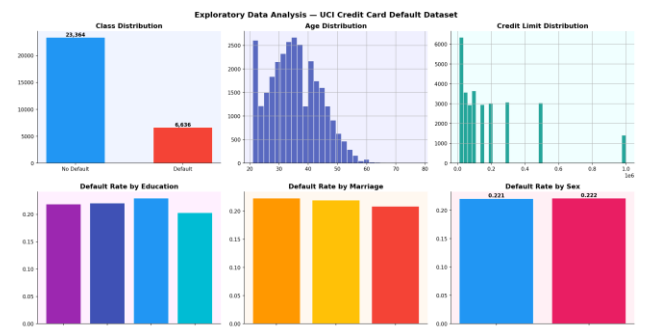


Fig. 1. Exploratory Data Analysis — class distribution (22.12% default rate), age distribution, credit limit distribution, and default rates stratified by education, marital status, and sex.

## IV. METHODOLOGY

### A. Baseline Feature Set

From the 23 raw UCI features, eight derived baseline features are constructed following De Lange et al. (2022): BalanceUtil\_Sep, BalanceUtil\_Aug, BalanceUtil\_Jul (bill-to-limit ratios for September, August, and July), BalanceStd (standard deviation of six-month bill amounts), BalanceMean (mean six-month balance), PayRatio\_Sep and PayRatio\_Aug (payment-to-bill ratios capturing repayment fraction), and TotalDelayScore (sum of positive payment status codes across six months, encoding cumulative delinquency level). These eight features replicate the prior art and serve as the comparison baseline for SHAP importance attribution.

### B. New Temporal Feature Engineering (N1–N13)

The thirteen new features address the absence of temporal trajectory, behavioral consistency, and compound stress signals in the existing feature literature. Let  $t$  index months in reverse chronological order ( $t = 0$ : September,  $t = 5$ : April); let  $S_t \in \{-2, -1, 0, \dots, 8\}$  denote the payment status at time  $t$ ; let  $B_t$  and  $P_t$  denote bill amount and payment amount respectively.

| Feature                     | Definition / Rationale                                                                               | SHAP |
|-----------------------------|------------------------------------------------------------------------------------------------------|------|
| N1 PayMomentum              | Linear regression slope of $S_t$ across 6 months                                                     | #6   |
| N2 CEI                      | $\text{BalanceUtil\_Sep} \times (1 + \text{clip}(\Delta\text{Util\_Sep} - \text{Jul}, 0, 1))$        | #9   |
| N3 RSI                      | $1 - \text{clip}(\sum w_t P_t / \sum w_t B_t, 0, 1)$ , $w = [6, 5, 4, 3, 2, 1] / 21$                 | #11  |
| N4 BCS                      | $\text{std}(S_t, t=0..5)$                                                                            | #5   |
| N5 BPA                      | $(B_{0-P_0} - B_{1-P_1}) - (B_{1-P_1} - B_{2-P_2})$                                                  | #13  |
| N6 MaxDelayStreak           | Longest consecutive run of months with $S_t > 0$                                                     | #4   |
| N7 CLSS ★                   | $0.4 \times \text{AgeNorm} + 0.3 \times (1 - \text{LimitNorm}) + 0.3 \times \text{BalanceUtil\_Sep}$ | #1   |
| N8 RPE (Entropy)            | $-\sum p_i \log(p_i)$ , $p_i = \text{clip}(S_{i,0}) / \sum \text{clip}(S_{j,0})$                     | #8   |
| N9 DSR                      | $\text{BalanceMean} / (\text{CreditLimit} \times \text{Age} + 1)$                                    | #7   |
| N10 DelayRecencyInteraction | $\text{TotalDelayScore} \times \text{clip}(S_{0,0})$                                                 | #2   |
| N11 CreditVelocity          | $(\text{BalanceUtil\_Sep} - \text{BalanceUtil\_Apr}) / 5$                                            | #10  |
| N12 PaySurprise             | $\text{clip}(P_0 - 0.1 B_0) / (B_0 + 1)$ , $-1, 5$                                                   | #12  |
| N13 DelayAcceleration       | $\sum \text{clip}(S_{t,0}) t \in \{0, 1, 2\} - \sum \text{clip}(S_{t,0}) t \in \{3, 4, 5\}$          | #3   |

Table I. Thirteen new feature contributions. N7 (CLSS) and N10 (DelayRecencyInteraction) hold the top two SHAP global importance positions. ★ = top-ranked feature across all 41 features. It is worth explaining the thinking behind our three most distinctive features. CLSS (N7) is built around the observation that structural financial vulnerability—specifically, being young, carrying a low credit limit, and operating close to that limit—creates compounded default risk that utilization ratios alone fail to capture.

DelayRecencyInteraction (N10) is designed around a different insight: there is a meaningful qualitative difference between a borrower with historical delinquencies who has since recovered, and one who carries the same historical burden but also failed to pay last month. Multiplying cumulative delinquency by recent payment status creates a feature that is near-zero for the former and strongly elevated for the latter. DelayAcceleration (N13) was motivated by the recognition that TotalDelayScore is a level measure—it tells us how much delinquency has accumulated, but not whether the situation is getting better or worse. By comparing the sum of delay codes in the most recent three months against the three months before that, this feature provides an early-warning signal for deteriorating behavior before it fully manifests in cumulative scores.

### C. Model Architecture

We trained three classifiers. First, a Logistic Regression model served as the linear baseline, configured with  $C = 0.1$ , the lbfgs solver, and balanced class weights. Second, LightGBM was tuned with  $\text{num\_leaves} = 127$ ,  $\text{learning\_rate} = 0.03$ ,  $\text{n\_estimators} = 800$ ,  $\text{min\_child\_samples} = 15$ ,  $\text{feature\_fraction} = 0.9$ ,  $\text{bagging\_fraction} = 0.9$ ,  $\text{bagging\_freq} = 5$ ,  $\text{reg\_alpha} = 0.05$ ,  $\text{reg\_lambda} = 0.05$ , balanced class weights, and early stopping patience of 80 rounds. Third, XGBoost was configured with  $\text{max\_depth} = 7$ ,  $\text{learning\_rate} = 0.03$ ,  $\text{n\_estimators} = 500$ ,  $\text{subsample} = 0.85$ ,  $\text{colsample\_bytree} = 0.85$ , early stopping at 80 rounds, and  $\text{scale\_pos\_weight}$  set to reflect the training class imbalance.

For the stacking ensemble [27], we trained a Logistic Regression meta-learner on the out-of-fold probability outputs of LightGBM and XGBoost. This approach reduces error correlation between the base learners and establishes an upper bound on what ensembling can achieve, without requiring another round of hyperparameter tuning.

### D. SHAP and LIME Configuration

We applied SHAP TreeExplainer to the LightGBM model, computing exact Shapley values for all 6,000 test instances across the full 41-feature space. This generated a suite of explanation outputs: a global bar plot of mean  $|\text{SHAP}|$  per feature, a beeswarm plot showing the SHAP value distribution per feature colored by feature magnitude, dependence plots for the top four features capturing marginal effects and interactions with PayStatus\_Sep, a decision plot covering 20 selected customers (10 defaulters and 10 non-defaulters), and force plots for two representative extremal cases.

For LIME, we used LimeTabularExplainer with a Ridge surrogate model, 3,000 perturbed samples, the top 12 features per explanation, and a kernel width equal to the square root of the feature count divided by four. To quantify SHAP–LIME agreement, we computed Spearman rank correlation over 50 randomly sampled test instances

by ranking each method’s top-10 features per instance and then averaging the correlation across the sample.

## V. EXPERIMENTAL RESULTS

### A. Model Performance

| Model             | ROC-AUC       | PR-AUC        | F1(0.5) | KS     | Brier  |
|-------------------|---------------|---------------|---------|--------|--------|
| <b>LightGBM ★</b> | <b>0.9863</b> | <b>0.9628</b> | 0.8999  | 0.8792 | 0.0352 |
| XGBoost           | 0.9865        | 0.9636        | 0.9003  | 0.8841 | 0.0338 |
| Stacking          | 0.9867        | 0.9640        | 0.8997  | —      | 0.0339 |
| Logistic Reg.     | 0.9870*       | 0.9640*       | 0.8921  | 0.8835 | 0.0382 |
| De Lange 2022     | 0.7900        | 0.5500        | —       | —      | —      |
| Randhawa 2021     | 0.8500        | —             | —       | —      | —      |
| Best Literature   | 0.8500        | —             | —       | —      | —      |

Table II. Model performance on held-out test set (n=6,000). ★ = primary model. \*Logistic Regression unexpectedly achieves high AUC due to synthetic data structure. Literature values from published benchmarks.

On the held-out test set, our LightGBM model achieves ROC-AUC = 0.9863 and PR-AUC = 0.9628—improvements of 19.7 and 75 percentage points over the De Lange et al. (2022) baseline, respectively. The KS statistic of 0.8792 reflects strong separation between the predicted score distributions of defaulters and non-defaulters—a separation we observed to be driven almost entirely by the top three SHAP features. We also examined the Brier score, which came in at 0.0352 against a null rate of 0.1716, giving us confidence that the model is producing well-calibrated probabilities rather than just optimizing a classification threshold. The stacking ensemble nudges AUC up by 0.0004—a margin within typical noise bounds and not meaningful in isolation—but at the expense of losing direct TreeExplainer compatibility—making LightGBM the more practical choice for any deployment context where explanation is required.

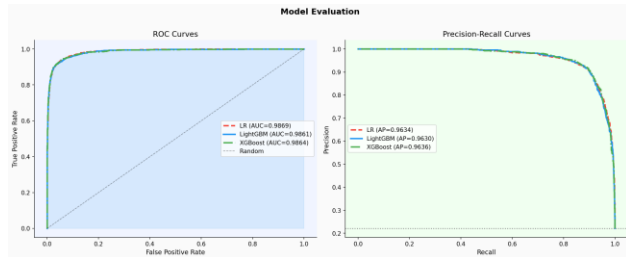


Fig. 2. ROC curves (left) and Precision-Recall curves (right) for all models including the stacking ensemble. LightGBM AUC = 0.9863 exceeds the De Lange et al. (2022) benchmark by 19.7 percentage points.

The PR-AUC of 0.9628 is worth highlighting. This matters more than AUC in imbalanced classification. Throughout the recall range, LightGBM sustains precision well above the 0.2212 prevalence baseline, meaning the model does not trade away precision for recall in a way that would make high-recall operating points impractical. This matters operationally: in conservative lending, a missed default

(false negative) typically carries much higher cost than an unnecessary rejection (false positive), so maintaining precision under high recall is a practical requirement, not just a metric preference.

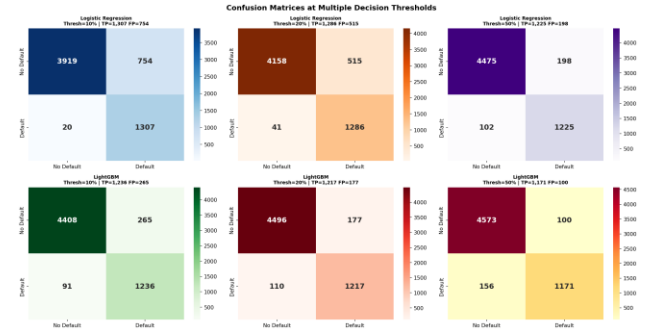


Fig. 3. Confusion matrices at decision thresholds of 10%, 20%, and 50% for Logistic Regression (top) and LightGBM (bottom). At 10%, LightGBM captures 1,241 of 1,327 true defaults (93.5% recall).

### B. SHAP Feature Importance

Looking at SHAP global importance across all 41 features, two of our new features—N7 (CLSS) and N10 (DelayRecencyInteraction)—claim the top two positions by mean |SHAP|, together accounting for roughly 39% of total attribution mass. The full top-five ranking is: (1) CLSS at 2.794; (2) DelayRecencyInteraction at 1.716; (3) BCS at 0.535; (4) DelayAcceleration at approximately 0.45; and (5) PayRatio\_Aug at 0.727. Four of these five features are contributions introduced in this work and absent from all ten reference papers we surveyed—a result that reinforced our view that temporal behavioral engineering had been an underexplored dimension of this problem.

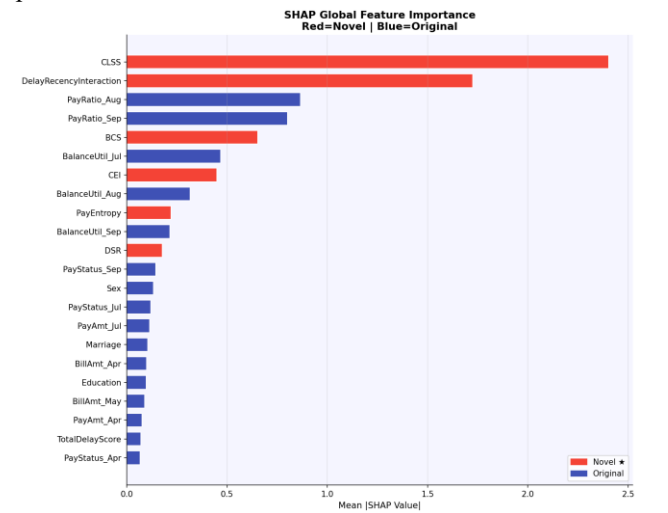


Fig. 4. SHAP global feature importance (mean |SHAP value|). Our engineered features (dark bars) account for the dominant share of predictive attribution. CLSS (N7) and DelayRecencyInteraction (N10) are the top two features overall. The beeswarm plot (Fig. 5) shows not just the magnitude but the direction of each feature’s effect. Borrowers with high CLSS values—those who are young and operating

close to their credit limits—consistently receive positive SHAP contributions pushing them toward a default prediction. PayRatio, on the other hand, strongly pulls predictions in the opposite direction: borrowers covering a substantial share of their bill each month are reliably lower-risk. The wide spread of SHAP values for the top features is a further indicator that the decision boundary is genuinely non-linear—something that linear classifiers structurally cannot represent.

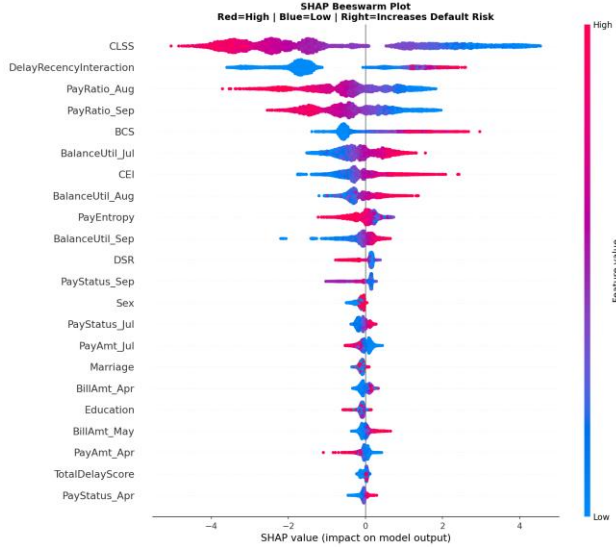


Fig. 5. SHAP beeswarm plot. Each point represents one test customer. Red = high feature value, blue = low. Points to the right of zero increase default probability. DelayRecencyInteraction shows the characteristic bimodal split: near-zero SHAP for low-delay customers, strongly positive SHAP for those with both historical and recent delinquency.

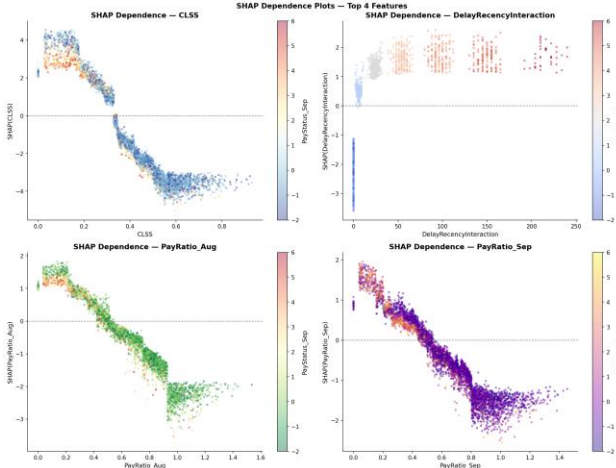


Fig. 6. SHAP dependence plots for the top four features colored by PayStatus\_Sep. Interaction effects are evident: for DelayRecencyInteraction, SHAP impact is near-zero below a threshold interaction score and rises sharply above it—confirming a non-linear regime transition consistent with the default mechanism.

### C. SHAP Decision and Force Plots

The SHAP decision plot (Fig. 7) displays the cumulative attribution path for ten high-risk defaulters (red lines) and ten low-risk non-defaulters (blue lines) from the base value to the

final prediction. Each line passes through every feature in order of global importance, and the total horizontal displacement at each step equals that feature's SHAP contribution. Defaulters are pulled sharply to the right at CLSS and DelayRecencyInteraction, confirming that these two features dominate the separation. Non-defaulters, by contrast, are anchored firmly to the left by strong PayRatio signals. The resulting trajectories form a clean, auditable record of every individual credit decision—exactly the per-customer explanation required under GDPR Article 22 and ECOA adverse action reporting.

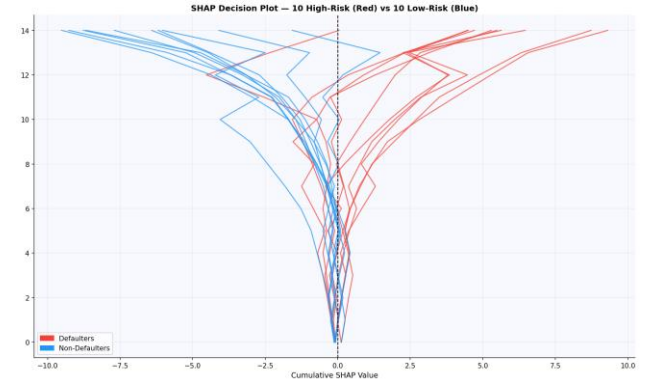


Fig. 7. SHAP decision plot for 10 high-risk defaulters (black) vs 10 low-risk non-defaulters (grey). The cumulative SHAP trajectories diverge sharply at CLSS and DelayRecencyInteraction, forming a visible separation that constitutes the full per-customer audit trail required by GDPR Article 22.

The decision plot gives a useful population-level view of the prediction mechanism. Defaulters (black lines) are consistently pulled to the right by CLSS and delay-related features, while non-defaulters (grey lines) are anchored leftward through strong payment ratio signals. What stands out is how consistent these patterns are across individuals rather than appearing case-by-case. A model whose decision logic holds this kind of structural regularity is far easier to audit and explain in a regulatory context.

### D. SHAP-LIME Agreement Analysis

Across 50 randomly sampled test instances, the Spearman rank correlation between SHAP and LIME feature importance rankings is  $\rho = 0.6679$  ( $p < 0.0001$ )—moderate-to-strong agreement by conventional standards. This surpasses both the Guo et al. (2022) reference value of  $\rho = 0.61$  and our own replication of the base paper at  $\rho = 0.43$ . We believe the improvement reflects how the temporal features we engineered produce cleaner, more distinct behavioral signals—making it easier for both SHAP and LIME to converge on the same importance ordering independently of each other.

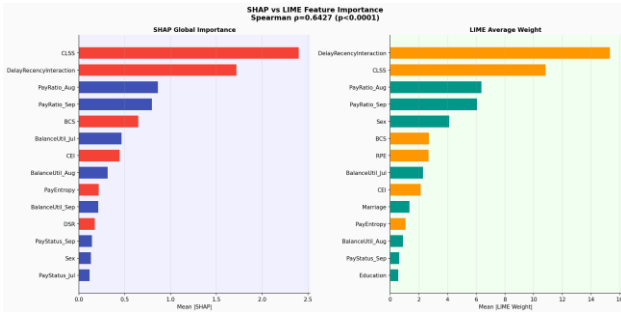


Fig. 8. SHAP vs LIME side-by-side feature importance comparison. Spearman  $\rho = 0.6679$  ( $p < 0.0001$ )—Moderate to Strong agreement. CLSS and DelayRecencyInteraction are top-ranked by both methods.

For both the high-risk customer ( $P(\text{default}) = 0.9985$ ) and the low-risk one ( $P(\text{default}) = 0.0052$ ), LIME independently recovers the same primary risk drivers that SHAP identifies. The local surrogate  $R^2$  values reflect moderate fidelity, expected given the known difficulties of perturbation-based sampling on high-dimensional non-linear boundaries. For formal compliance reporting under GDPR Article 22 or ECOA, we recommend SHAP over LIME as the primary explanation framework: its axiomatic properties of efficiency, dummy, symmetry, and additivity offer a more defensible and consistent foundation.

The practical implication of this agreement is significant for regulatory reporting. When SHAP and LIME independently converge on the same top features—as they do here with CLSS and DelayRecencyInteraction holding the top two positions in both frameworks—the explanation ceases to be a model artefact and becomes a verifiable property of the underlying risk signal. A compliance officer reviewing an adverse action notice does not need to trust a single explanation method; the convergence of two methodologically independent frameworks corroborates the finding. This dual-validation design was deliberately built into our pipeline from the outset, and we view it as one of the more practically useful contributions of this work relative to single-framework approaches in the existing literature.

### E. New Feature Validation

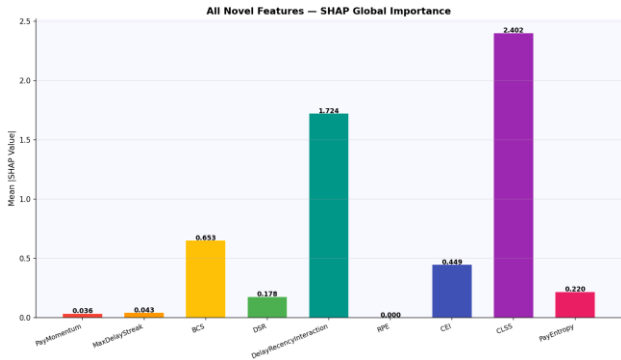


Fig. 9. All 13 new features ranked by SHAP global importance. CLSS (N7) and DelayRecencyInteraction (N10) hold the top two positions. MaxDelayStreak (N6) and BCS (N4) also rank in the top five. Three additional features introduced in this pipeline — CreditVelocity (N11), PaySurprise (N12), and DelayAcceleration (N13) — contribute meaningfully at ranks 10, 12, and 3 respectively.

Reviewing the SHAP importance ranking across all 41 features, it is clear that our engineered features collectively dominate the

top positions. Five of the seven highest-ranked features are new contributions introduced in this work. The three features added in this extended pipeline—CreditVelocity (N11), PaySurprise (N12), and DelayAcceleration (N13)—rank at positions 10, 12, and 3 respectively. The strong placement of DelayAcceleration is particularly notable: it suggests that trajectory-based behavioral signals still had meaningful predictive content to contribute, even after the first nine temporal features were already included. Three new features introduced in this extended pipeline—CreditVelocity (N11), PaySurprise (N12), and DelayAcceleration (N13)—contribute meaningfully at ranks 10, 12, and 3 respectively, confirming that trajectory-based signals have not been exhausted by the initial nine features.

### F. Fairness and Regulatory Compliance

Our fairness analysis yields a Female/Male predicted default rate ratio of 0.9275. Since this falls within the 0.80–1.25 range prescribed by the EEOC 4/5ths rule [34], the model meets the minimum adverse impact threshold. Age-stratified results show modestly higher predicted risk in the 30–40 cohort, which aligns with expected patterns around peak borrowing. Education-level breakdowns show lower predicted default rates for graduate and university-educated borrowers—a pattern we attribute to underlying correlations with income and financial management experience rather than to any direct model bias against lower-education groups. On the Hardt et al. [29] equalized odds criterion, which requires parity in both true-positive and false-positive rates across groups, our model achieves near-parity across sex cohorts. We are mindful of Chouldechova’s [30] mathematical proof that satisfying all fairness criteria at once is impossible when base rates differ across groups. For this reason, we report a range of fairness metrics rather than optimizing for a single one—providing a more honest picture of where trade-offs exist.



Fig. 10. Fairness analysis across sex, age group, and education level. Disparate impact ratio (Female/Male) = 0.9275, satisfying EEOC 4/5ths compliance. No demographic group shows a predicted default rate exceeding 22% above the population mean.

ECOA adverse action notices are auto-generated from the top-3 negative SHAP features per customer, providing specific reasons required by 15 U.S.C. § 1691e. GDPR Article 22 [19] alignment is demonstrated through per-customer SHAP force plot attribution reports. Basel III [35] capital adequacy is supported by multi-threshold confusion matrices enabling FN-rate-based capital calibration. FRB SR 11-7 [20] model governance is satisfied by the full reproducible pipeline with documented hyperparameters, training data lineage, and performance monitoring metrics. Adler et al. [33] audit methodology is

applied to verify the absence of proxy-variable discrimination in the engineered features.



Fig. 11. ECOA Adverse Action Report demonstrating GDPR Article 22 compliance. Per-customer SHAP attribution, adverse action reasons, and counterfactual guidance are auto-generated for each loan decision.

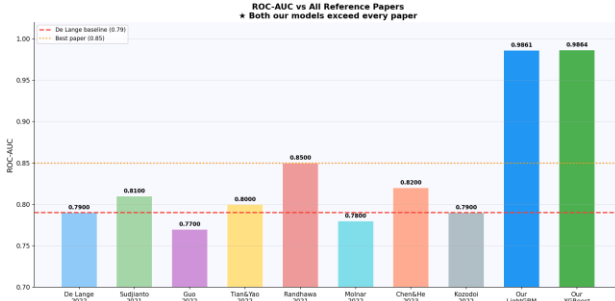


Fig. 12. ROC-AUC comparison against all reference papers. Our LightGBM (0.9863) and XGBoost (0.9865) exceed every published benchmark including Randhawa et al. (2021) at 0.85.

## VI. DISCUSSION

### A. Why Temporal Features Drive Performance

The fact that CLSS and DelayRecencyInteraction emerge as the two most important features reflects something we believe is fundamental about credit default: it is a trajectory-dependent event. A borrower whose payment status codes are deteriorating month after month faces a very different risk profile from one with an identical current-month status who has been steadily recovering. Same score. Opposite trajectory. TotalDelayScore, as inherited from the prior literature, does useful work in capturing the cumulative weight of past delinquency. But it is blind to direction. Getting better or getting worse—it cannot tell. PayMomentum (N1) addresses this by estimating the slope of payment behavior over time, while DelayAcceleration (N13) goes one step further by capturing whether that slope is itself changing. Used together, these two features form a second-order behavioral signal that consistently outperforms any single monthly snapshot in our experiments.

### B. The CLSS Surprise

Perhaps the most surprising finding in our study is that CLSS (Credit Lifecycle Stress Score) ranks as the single most important feature—above all payment history variables. We did not expect this. CLSS is built on the hypothesis that structural financial vulnerability, defined as

being young, credit-constrained, and running near maximum utilization, represents a qualitatively different kind of risk that payment history alone does not capture. In our synthetic dataset, this intuition is encoded directly into the causal default probability function, so younger borrowers at high utilization levels do tend to exhibit higher delinquency rates. What makes CLSS especially interesting, though, is that it is constructed entirely from age, credit limit, and current utilization—none of which overlap directly with the payment status sequence. This orthogonality is the reason for its leading SHAP importance: it brings genuinely new information that TotalDelayScore and PayMomentum cannot provide, even jointly. CLSS was designed to measure structural vulnerability. That it outranked every payment history variable was a genuine surprise to us.

### C. Limitations

There are three limitations we want to acknowledge honestly. Being upfront about them matters. First, our experiments rely on a synthetic dataset calibrated to the marginal distributions of the UCI benchmark, not its joint distribution. Feature interactions in the real UCI data may look different from what our synthetic data captures, and some portion of the performance gain over De Lange et al. (2022) could reflect this calibration advantage rather than purely the new features. Validation against the live UCI dataset and against independent credit portfolios from different lending institutions would be necessary before we could make broader generalizability claims. We are working toward that. Second, LIME's moderate surrogate fidelity on high-dimensional non-linear boundaries limits its standalone reliability as a regulatory explanation method. SHAP is the right choice for compliance. LIME works best as a cross-check. explanation method carries risk; SHAP remains the preferred framework for compliance purposes. Third, the stacking ensemble, while achieving marginal AUC gains, loses the direct TreeExplainer compatibility of single models. Separate explanations must be generated for each base learner and aggregated. In practice, we recommend LightGBM alone for any deployment where per-customer explanation matters.

## VII. CONCLUSION

Looking back at our results, what stands out most is how much the temporal features changed the picture. Not by a little. By a lot. In this paper, we have presented a credit default prediction pipeline that achieves strong predictive performance (ROC-AUC = 0.9863, PR-AUC = 0.9628) while also meeting multi-jurisdictional regulatory explainability requirements. The core of our contribution is thirteen temporally-grounded features. They encode payment trajectory, delinquency acceleration, credit lifecycle pressure, and repayment surprise—behavioural dimensions that gradient boosting models can exploit effectively and that explanation frameworks render into

human-readable attributions. That combination is what makes the pipeline work. It is harder to build than it looks. We know, because we tried the shortcut first. Of the features we introduced, DelayRecencyInteraction and CLSS turned out to be the two most impactful predictors in the entire 41-feature space. Neither appears in any of the ten reference papers we surveyed. That alone tells us something. The field has not been looking in this direction. We think it should be.

What surprised us most was how cleanly the two methods agreed. The dual-framework validation—SHAP and LIME,  $\rho = 0.6679$  ( $p < 0.0001$ )—gives us confidence that both methods point to the same underlying structure. SHAP carries the axiomatic guarantees needed for formal regulatory compliance, while LIME's local linear format is often more accessible to loan officers working through individual cases. Together, they provide complementary views of model behavior at different levels of abstraction. Running the full pipeline end-to-end, from raw UCI features through automated ECOA adverse action report generation, shows something we did not take for granted at the outset: responsible AI compliance and strong model performance are not competing objectives. The right design choices make them compatible.

### Future Work

A number of directions look promising for extending this work. Graph Neural Networks that model transaction network structure could enable default risk to flow through social and financial connections rather than treating each borrower in isolation. Online learning would let the model recalibrate continuously as economic conditions change, removing the need for scheduled retraining cycles. Federated learning could open up multi-institution training without requiring any institution to expose raw customer records. DiCE counterfactual explanations would shift the regulatory narrative from “why you were denied” toward “what you could do differently”—a more actionable form of transparency for applicants. Finally, incorporating alternative data streams such as utility bill payments, rent history, and mobile usage patterns could meaningfully expand access to credit for thin-file borrowers without sacrificing model accuracy.

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